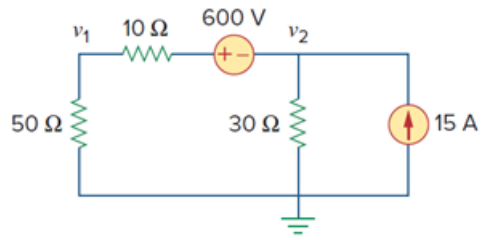


Problem

Calculate v_1 and v_2 in the circuit of Fig. 3.62 using nodal analysis.

Figure 3.62:

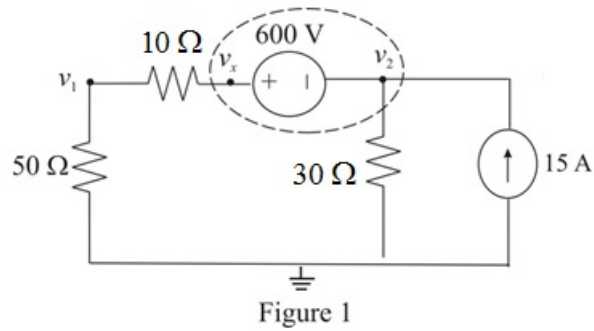


Step-by-step solution

Step 1 of 5

Refer to Figure 3.62 in the text book.

Identify the voltages in the circuit shown in Figure 1.



Step 2 of 5

Consider the relation between the voltages v_x and v_2 .

$$v_x - v_2 = 600$$

$$v_x = v_2 + 600 \quad \dots\dots (1)$$

Therefore, the expression for the voltage v_x is, $v_2 + 600$.

Step 3 of 5

Apply super node analysis at nodes 2 and x.

$$\frac{v_2}{30} + \frac{v_x}{10+50} - 15 = 0$$

$$\frac{v_2}{30} + \frac{v_x}{60} - 15 = 0$$

$$\frac{2v_2 + v_x - 900}{60} = 0$$

$$2v_2 + v_x - 900 = 0$$

Substitute $v_2 + 600$ for v_x .

$$2v_2 + v_2 + 600 - 900 = 0$$

$$3v_2 - 300 = 0$$

$$3v_2 = 300$$

$$v_2 = \frac{300}{3}$$

$$v_2 = 100 \text{ V}$$

Therefore, the value of the voltage v_2 is, 100 V.

Step 4 of 5

Recall equation (1).

$$v_x = v_2 + 600$$

Substitute 100 V for v_2 .

$$\begin{aligned} v_x &= 100 + 600 \\ &= 700 \text{ V} \end{aligned}$$

Therefore, the value of the voltage v_x is, 700 V .

Step 5 of 5

Apply node analysis at node 1.

$$\frac{v_1}{50} + \frac{v_1 - v_x}{10} = 0$$

$$v_1 \left(\frac{1}{50} + \frac{1}{10} \right) = \frac{v_x}{10}$$

$$v_1 \left(\frac{1+5}{50} \right) = \frac{v_x}{10}$$

$$v_1 = \frac{50}{60} v_x$$

Substitute 700 V for v_x .

$$\begin{aligned} v_1 &= \left(\frac{50}{60} \right) (700) \\ &= (0.8333)(700) \\ &= 583.3 \text{ V} \end{aligned}$$

Therefore, the value of the voltage v_1 is, 583.3 V.